

Final Project

Margarida Lupi Correia (51929), Samuele Siano (59546), Giada Tonietto (59547), Andjela Seatovic (64571)

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1. Introduction

In an era of increased market uncertainty and algorithmic trading, systematic strategies that exploit the statistical properties of asset price dynamics across time have gained renewed attention among academics and practitioners. Among these, the Time Series Momentum (TSMOM) framework, as introduced by Moskowitz, Ooi, and Pedersen (2011)[1], has emerged as a foundational model in the study of trend-following behavior and its implications for asset pricing. Unlike its cross-sectional counterpart, TSMOM is purely intratemporal: it builds directional positions based on an asset's own return history, entirely agnostic to the performance of its peers. This self-contained logic makes the strategy appealing not only for its empirical tractability but also for its theoretical alignment with behavioral underreaction, delayed information diffusion, and structural inertia in price formation. This aligns with a broader trend in financial research that increasingly embraces frameworks which challenge traditional assumptions of market efficiency, investor rationality, and frictionless trading, highlighting the growing relevance of behavioral and structural perspectives in understanding asset price dynamics.

Despite its empirical success across asset classes and geographies, the TSMOM framework is typically evaluated through static lenses (e.g., performance metrics, Sharpe ratios, or factor regressions). What remains underexplored is how the strategy behaves under macro-financial stress. In particular, little is known about how the strategy responds to abrupt regime shifts in market conditions, such as those induced by surges in implied volatility, policy uncertainty, or liquidity dislocations. This limitation is especially relevant given TSMOM's backward-looking nature: it adjusts its positions based on lagged return patterns, rather than contemporaneous macro signals. As such, understanding how the strategy reacts - or fails to react - in the presence of shocks becomes essential to evaluate its robustness and viability in real-world portfolios.

2. Objective and Methodology

In this paper, we address this gap by proposing a two-pronged analysis. First, we reproduce the core findings of Moskowitz, Ooi, and Pedersen (2011) on Time Series Momentum (TSMOM) across a diversified universe of futures contracts spanning four asset classes (commodities, equities, sovereign bonds, and currencies) using updated data, from 1988 to 2023. Our implementation follows the precise specifications of the original model and extends its empirical coverage to include the post-crisis and post-pandemic periods, both marked by extreme fluctuations in volatility and correlation structures. Afterwards, we extend the analysis by investigating how TSMOM returns interact with macro-financial variables, by introducing a structural econometric extension via a bivariate Vector Autoregressive (VAR) model, wherein we study the endogenous interaction between the TSMOM portfolio returns and log-transformed implied volatility (logVIX). This framework enables us to derive Impulse Response Functions (IRFs) that quantify the dynamic response of the strategy to volatility shocks, shedding light on its capacity for regime adaptation, signal degradation, or structural resilience.

The core contribution of our work lies in shifting the analysis of TSMOM from a purely statistical exercise to one rooted in macro-financial dynamics. By treating the strategy itself as an economic object - one whose returns evolve under the pressure of latent risk regimes - we provide a new interpretive lens through which to understand momentum as both a source of return and a function of informational friction.

3. Summary of the paper analyzed

In their influential 2011 paper, Moskowitz, Ooi, and Pedersen take a fresh look at momentum investing, not by comparing assets to each other as in the Cross-Sectional Momentum 1993 paper from Jegadeesh & Titman, but by focusing on how each asset behaves over time. The strategy exploits the empirical observation that past returns contain predictive power for future returns, not through cross-asset comparison, but through intratemporal autocorrelation patterns within each asset's own price process. It suggests that an asset that has been going up over the past year is more likely to keep going up in the near future. If it has been falling, it is more likely to keep falling. This approach, which they call Time Series Momentum (TSMOM), builds trading signals based solely on an asset's own past returns.

What makes their study stand out is how broad and consistent the results are. They test the strategy across 58 liquid futures and forward contracts across four major asset classes - equities, bonds, commodities, and currencies - and find that TSMOM works almost everywhere. A diversified TSMOM portfolio delivers strong, positive returns, with low correlation to traditional risk factors, high Sharpe ratios (often above 1) and performs especially well during periods of market stress. The authors also show that these trends tend to persist for up to a year, but then start to reverse, suggesting that markets underreact in the short term and overreact in the long term.

Moskowitz, Ooi, and Pedersen go a step further by exploring several theoretical mechanisms that could explain the

persistent profitability of TSMOM strategies. Rather than attributing the effect to a single anomaly, they suggest that TSMOM may arise from a combination of behavioral biases, like conservatism, herd behavior and anchoring, slow information diffusion and structural market frictions, like transaction costs, capital constraints, and risk management rules that limit the ability of arbitrageurs to counteract trends, all of which are consistent with a market environment where prices do not instantly reflect all available information. A particularly novel contribution of the paper is the use of Commodity Futures Trading Commission (CFTC) position data, to examine the behavior of different market participants. The authors find that speculators tend to follow trends, increasing their positions in the direction of recent returns, while hedgers often take the opposite side, likely due to their need to offset business risks rather than seek returns. This dynamic supports the hedging pressure hypothesis, where speculators are compensated for providing liquidity to hedgers, effectively earning a premium by riding the trend.

Overall, the paper challenges the idea that markets are fully efficient and that prices always reflect all available information. Instead, it shows that simple, systematic strategies like TSMOM can uncover persistent patterns in asset prices and that these patterns may be driven by real-world frictions, behavioral biases, and institutional constraints.

4. Data Collection and Preprocessing

To conduct our empirical replication and extension of the TSMOM strategy, we compiled a comprehensive dataset consisting of monthly closing prices for a broad universe of futures contracts across four asset classes: commodities, equity indices, sovereign bonds, and currencies. The data spans from 1988 to 2023 and was sourced using the Bloomberg Terminal. All data was retrieved in continuous format (back-adjusted futures), which ensured consistency across contract rollovers and avoided artificial jumps due to delivery dates. Additionally, we collected monthly closing values of the CBOE Volatility Index (VIX), over the full sample period, which serves as a proxy for implied market volatility. This series was later log-transformed to better capture proportional changes in market-implied volatility and to reduce the impact of scale differences in the VAR analysis.

Data extraction was performed via the Bloomberg Excel Add-In, using manually validated tickers corresponding to the instruments listed in the original paper (**Table 1**). Once the data was retrieved, we structured it into a panel format with a uniform monthly timestamps. All price series were converted to numeric format and aligned by date. We then applied a data-cleaning where instruments with insufficient historical coverage were excluded from the analysis and rows with missing values across all instruments were dropped. Specifically, we imposed a missing data threshold of 30%: any instrument exhibiting more than 30% missing observations over its potential sample was discarded. The same thresholding procedure was then applied to individual years with excessive data gaps. The resulting dataset corresponds to the filtered panel described above. Instruments removed during this cleaning stage are listed in **Table 2** for reference.

The final dataset is unbalanced in nature: the number of instruments included in the TSMOM portfolio varies across time depending on data availability. This structure closely replicates the empirical setup of Moskowitz, Ooi, and Pedersen (2011), while allowing for a cleaner integration into our Python-based estimation framework, suitable for both the replication of the TSMOM strategy and the subsequent econometric modeling.

5. Strategy Replication

Logarithmic returns were computed from price levels using the standard formula $r_t = \log(P_t/P_{t-1})$. These returns form the basis for the momentum signal, which evaluates the direction and strength of recent price trends for each asset individually. Formally, let $r_{i,t}$ denote the log return of asset i at time t . The TSMOM signal is computed as the sign of the cumulative return over a backward-looking window of k periods, excluding the current period. In our implementation, consistent with the original paper, we adopt $k = 12$, and construct the signal as:

$$\text{Signal}_{i,t} = \text{sign} \left(\sum_{j=1}^{12} r_{i,t-j} \right)$$

This signal takes on a value of +1 if the cumulative return is positive, indicating a long position; -1 if the return is negative, indicating a short position; and 0 if the return is zero, in which case the strategy remains neutral. This rule allows us to systematically identify and follow persistent trends in each asset's return series.

We then applied the signal to the contemporaneous return to compute the strategy return for each asset:

$$\text{StrategyReturn}_{i,t} = \text{Signal}_{i,t} \cdot r_{i,t}$$

The portfolio-level return at time t , denoted R_t^{TSMOM} , is computed as the equally weighted average of the individual strategy returns across the N_t instruments available at time t :

$$R_t^{\text{TSMOM}} = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{StrategyReturn}_{i,t}$$

Where N_t is time-varying due to data availability constraints. No volatility scaling or cross-sectional normalization is employed; each asset contributes equally in weight and is treated symmetrically with respect to its own lagged behavior.

The choice of a 12-month lookback window, as established in the literature, is not arbitrary. It balances the trade-off between responsiveness and signal strength: shorter windows tend to be more reactive but noisier, while longer horizons dilute information. The use of an equal-weighted aggregation across assets further ensures that the strategy’s performance is not driven by concentration in any single market, but instead reflects a systematic capture of persistence patterns across heterogeneous instruments.

6. Results and Performance

The Time Series Momentum (TSMOM) strategy implemented in this project, as described in the previous section, yielded the performance metrics summarized in **Table 3**.

These results are notably weaker than those reported in the original study by Moskowitz, Ooi, and Pedersen (2011), where the TSMOM strategy achieved Sharpe ratios greater than 1. As shown in **Figure 1**, the strategy failed to recover from the drawdown that began in 2009. Since then, it never reached the high returns it achieved earlier, remaining below its previous peak.

The results from the study show that the TSMOM portfolio consistently outperformed a diversified passive long-only portfolio across a range of asset classes. In contrast, the cumulative return plot from our implementation shows that TSMOM portfolio mostly underperforms the passive strategy from 2009 onwards (**Figure 2 and Figure 3**).

This discrepancy can be attributed to several important factors. Firstly, the period covered in this replication includes recent years characterized by heightened market volatility, as shown by the several peaks of the VIX index (**Figure 4**). Events such as the COVID-19 pandemic, inflation shocks, and geopolitical instability have led to frequent and severe spikes in volatility and shifts in market dynamics. Since the TSMOM strategy is inherently backward-looking, it reacts to past trends rather than anticipating abrupt changes. As a result, it often lags during times of high volatility and may be caught offside during rapid market reversals. This delayed response reduces the effectiveness of the strategy in turbulent markets.

Increased central bank intervention has altered market dynamics, like Quantitative Easing suppressing volatility. Low interest rate environments have impacted trend following effectiveness, particularly in bonds and currencies. Market liquidity and efficiency may have increased, making simple momentum less effective. The changing structure of financial markets could be a contributing factor to the underperformance observed in our replication.

Additionally, time series momentum has become a well-known and widely implemented approach among institutional investors and hedge funds. With greater awareness and adoption of this strategy, along with advances in algorithmic trading and faster information dissemination, market prices adjust more quickly to new information. This reduces the persistence of trends and compresses potential alpha, diminishing the edge that time series momentum strategies previously enjoyed.

Lastly, unlike the methodology used in the original paper, our implementation does not include volatility scaling. In the paper, each position was scaled to target a constant 40% annualized volatility using an ex-ante volatility estimate. One motivation for this choice is that unscaled signals provide a more transparent view of raw momentum effects, without introducing model-based assumptions about future volatility, which can be noisy or unreliable, especially during turbulent market periods. However, it is worth noting that not applying volatility scaling can result in greater influence from high-volatility assets, potentially increasing portfolio risk and lowering risk-adjusted returns.

In summary, while the implementation of the TSMOM strategy adheres to the core logic of the original study, the weaker performance likely stems from a combination of recent market volatility, the strategy’s lagging nature, the absence of volatility scaling, and fundamental shifts in financial markets over time. These findings highlight the need to continuously re-evaluate and adapt quantitative strategies to ensure their relevance and effectiveness under evolving market conditions. While “vanilla” TSMOM may struggle, more sophisticated or adaptive approaches may still work, as demonstrated by Lim, Zohren, and Roberts in their 2019 paper (*“Enhancing Time-Series momentum strategies using deep neural networks”*)[2], where they utilize Long Short-Term Memory (LSTM) networks to capture temporal dependencies in financial time series.

7. VAR Extensions

Vector Autoregression (VAR) is an econometric framework designed to capture the dynamic interdependencies among multiple time series variables. The core idea is straightforward: each variable in the system is modelled as a linear function of its own past values and the past values of all other variables included in the model. This symmetric structure treats all variables as potentially endogenous, without imposing any ex-ante theoretical hierarchy or causal ordering. The primary purpose of a VAR model is to study how unexpected shocks to one variable propagate through the system over time. This is accomplished through the computation of Impulse Response Functions (IRFs), which trace the temporal reaction of each variable to an isolated structural innovation. In the context of our analysis, the

VAR framework allows us to investigate how returns from the TSMOM strategy respond to a sudden increase in implied volatility, as captured by log-transformed VIX. What makes VAR particularly appealing is its flexibility: it does not rely on strict theoretical assumptions about the underlying relationships between variables but instead lets the data reveal the structure of temporal dependencies. This makes it especially useful in our setting, where the objective is to assess whether and how the TSMOM strategy absorbs information from volatility shocks, and to what extent this adjustment process unfolds gradually over time. By doing so, the VAR model offers an empirically grounded way to evaluate the strategy’s structural resilience and adaptive capacity in the presence of shifting risk regimes.

8. VAR Specification and Estimation

Before implementing the Vector Autoregressive (VAR) model, it is essential to ensure that its underlying assumptions are met. A VAR model assumes that the time series involved are stationary, meaning that their statistical properties - such as mean, variance, and autocovariance - do not change over time. In particular, the model requires covariance stationarity, which ensures that shocks dissipate rather than accumulate, allowing for meaningful interpretation of the estimated coefficients and impulse response functions.

Beyond stationarity, the VAR framework assumes the absence of serial correlation in the residuals and the stability of the system. Stability is verified by ensuring that all eigenvalues of the companion matrix lie strictly within the unit circle. This condition guarantees that the effect of shocks dies out over time and that the system does not explode dynamically.

To verify these assumptions, we first applied the Augmented Dickey-Fuller (ADF) test to each variable. As shown in **Table 4**, all relevant series - including TSMOM returns and logVIX - were found to be stationary at the 5 percent significance level. The TSMOM return series was found to be stationary in levels and was therefore included without further transformation, as it had already been log-transformed during the portfolio construction phase. The VIX series was log-transformed as well, both for consistency with the TSMOM specification and to improve interpretability. Differencing was tested but ultimately discarded, as it did not yield significant improvements in stationarity diagnostics.

Next, we selected the appropriate lag length using standard information criteria. Although the AIC and BIC suggested slightly different specifications, we relied on the Hannan–Quinn Information Criterion (HQIC), which indicated a lag length of 1. This criterion is widely regarded as a compromise between AIC’s tendency to overfit and BIC’s inclination to underfit, offering a balanced specification for relatively large samples like ours (**Table 5**).

The VAR(1) model was estimated using Ordinary Least Squares (OLS) (**Figure 5**), and its adequacy was assessed through post-estimation diagnostics regarding residual autocorrelation and stability.

We applied the multivariate version of the Breusch–Godfrey LM test to assess residual autocorrelation (**Table 6**). Although the LM test yielded a p-value marginally below the conventional 5 percent threshold, suggesting a mild presence of residual autocorrelation, we opted to retain the VAR(1) specification. This decision is grounded in several considerations. First, the detected autocorrelation is statistically mild and not indicative of strong misspecification. Second, higher-lag VAR models tend to suffer from overfitting, reduced degrees of freedom, and wider confidence intervals in finite samples - issues that can obscure rather than clarify dynamic relationships. Third, the HQIC criterion, which balances model fit and parsimony, clearly favors a one-lag specification. In line with standard practice in macro-financial applications, and given the overall stability and interpretability of the model, we judged the VAR(1) as a reasonable and robust approximation of the underlying data-generating process.

Lastly, stability was confirmed by checking the eigenvalues of the companion matrix (**Table 7**). All eigenvalues were strictly within the unit circle, confirming that the system satisfies the condition for dynamic stability and that the estimated VAR can be validly interpreted through impulse response analysis.

9. Impulse Response Function: Findings and Interpretation

The impulse response analysis offers a compelling window into the dynamic behavior of the TSMOM strategy in the presence of market stress. Specifically, we compute the IRF of TSMOM returns to a one-standard-deviation orthogonalized shock to logVIX. As shown in **Figure 6**, the resulting profile exhibits an initial negative reaction in the month immediately following the shock, followed by a gradual and monotonic reversion toward zero. The effect becomes negligible by the fourth month. The confidence band, while relatively wide, consistently straddles zero, making the response statistically non-significant at the 95% level, but its economic interpretation remains both clear and relevant.

To unpack the shape and meaning of this impulse response, it is essential to first understand what a shock to logVIX represents. Unlike shocks to prices or returns, which often imply directional movement, a positive innovation in logVIX is best interpreted as a jump in expected uncertainty. It does not convey new information about expected return levels but about the dispersion and volatility of future outcomes. In practical terms, this type of shock is tantamount to a regime change, one in which market participants suddenly expect a broader, more erratic distribution of future returns. Such shocks are commonly associated with macroeconomic dislocations, policy shifts, or abrupt repricings of risk premia.

The TSMOM strategy, being inherently backward-looking, constructs its signals based on realized past returns over a fixed window. When a volatility shock occurs, the strategy does not immediately recognize the change in regime. It continues to hold positions predicated on trends that developed under lower volatility conditions and that may no longer be valid. This creates a temporal mismatch between the signal and the underlying return-generating process. Consequently, the strategy's positions may be misaligned in the short term, leading to underperformance, which is precisely what we observe in the initial negative dip in the IRF.

However, the response is not persistently negative. As time passes, the new volatility regime is progressively incorporated into the trailing return window. The strategy begins to re-estimate its signals under the new distribution of returns, effectively recalibrating itself. This self-correcting mechanism explains the IRF's return toward zero. Importantly, this does not imply that the strategy hedges against volatility shocks; rather, it absorbs them over time via structural memory embedded in its signal-generation process. This reflects a slow-moving yet adaptive architecture that aligns with the economic intuition of trend-following in uncertain environments.

The presence of both negative and positive areas within the IRF's confidence band is equally telling. The wide bounds do not suggest ambiguity in the direction of the effect, but rather reflect the inherent volatility of strategy returns under stress. Momentum strategies are particularly exposed to reversals, noise, and discontinuities in return processes, all of which are more frequent during periods of elevated VIX. As such, the IRF's confidence interval naturally expands, encompassing the possibility of mild positive corrections even while the average response remains negative. It is essential to recognize that the IRF represents a distribution of possible responses, not a deterministic path. The fact that the mean response is negative, coupled with an early recovery, indicates a consistent but moderate vulnerability to sentiment shocks, followed by effective adaptation.

This impulse pattern is not only economically intuitive, but also structurally consistent with the internal logic of the TSMOM strategy. The strategy is not built to predict macro shocks ex-ante, but to track and react to persistent return patterns. When these patterns are disrupted by uncertainty shocks, a transient misalignment occurs. Yet, unlike strategies that overreact or amplify volatility (such as highly leveraged carry trades), TSMOM behaves more like a stabilizer: its response is muted, self-reverting, and informative of the market's new equilibrium. In this sense, the IRF confirms that TSMOM is not fragile in the face of volatility - it is deliberately slow-reacting, and this inertia becomes an asset once the strategy has re-learned the environment.

Final Thoughts and Further Research

This project set out to revisit the Time Series Momentum (TSMOM) strategy not merely as a rule-based trading heuristic, but as a dynamic system whose behavior reflects the underlying structure of macro-financial regimes. Through a rigorous empirical replication of Moskowitz, Ooi, and Pedersen (2011) and the integration of a bivariate VAR model linking strategy returns with implied volatility, we sought to understand how TSMOM behaves in the presence of volatility shocks and regime changes.

What emerges is a portrait of TSMOM as a self-learning system - one that does not forecast, but instead gradually adapts to its environment through endogenous signal recalibration. The impulse response functions (IRFs) illustrate this dynamic clearly: an initial negative reaction to a volatility shock, consistent with the strategy's lagged nature, is followed by a steady reversion as new market conditions are internalized. This adaptive mechanism is subtle but powerful, allowing the strategy to maintain functionality - even effectiveness - under structurally shifting risk conditions.

Rather than challenging the validity of the strategy, this pattern of response reinforces its robustness. TSMOM does not attempt to anticipate breaks ex-ante; instead, it absorbs them ex-post through systematic learning. This delayed, yet disciplined adjustment highlights the strategy's inherent resilience and justifies its persistent appeal in both academic research and professional asset management.

Nonetheless, our VAR specification revealed a mild presence of residual autocorrelation, suggesting that the chosen lag structure, while parsimonious and theoretically justified, may not fully capture the dynamics of the system. While this did not invalidate the model - given the stability of the companion matrix and the overall interpretability of the IRFs - it points to a potential avenue for refinement. Future iterations could explore higher-lag specifications or alternative modeling approaches, such as VARMA or structural VARs (SVARs), which may better accommodate the memory structure of financial time series without sacrificing parsimony.

In conclusion, our findings suggest that time series momentum remains a viable and informative strategy, but one whose effectiveness depends critically on its ability to adapt - not just to past returns, but to the volatility regime in which those returns unfold. Capturing this interaction is essential for understanding both the power and the limits of trend-following in today's complex market environments.

Appendix

Asset Class	Instrument
Commodity futures	ALUMINUM
Commodity futures	BRENT OIL
Commodity futures	CATTLE
Commodity futures	COCOA
Commodity futures	COFFEE
Commodity futures	COPPER
Commodity futures	CORN
Commodity futures	COTTON
Commodity futures	CRUDE
Commodity futures	GASOIL
Commodity futures	GOLD
Commodity futures	HEATOIL
Commodity futures	HOGS
Commodity futures	NATGAS
Commodity futures	NICKEL
Commodity futures	PLATINUM
Commodity futures	SILVER
Commodity futures	SOYBEANS
Commodity futures	SOYMEAL
Commodity futures	SOYOIL
Commodity futures	SUGAR
Commodity futures	UNLEADED
Commodity futures	WHEAT
Commodity futures	ZINC
Equity index futures	ASX SPI 200
Equity index futures	DAX
Equity index futures	IBEX 35
Equity index futures	CAC 40
Equity index futures	FTSE/MIB
Equity index futures	TOPIX
Equity index futures	AEX
Equity index futures	FTSE 100
Equity index futures	S&P 500
Bond futures	3-year AUS
Bond futures	10-year AUS
Bond futures	2-year EURO
Bond futures	5-year EURO
Bond futures	10-year EURO
Bond futures	30-year EURO
Bond futures	10-year CAN
Bond futures	10-year JP
Bond futures	10-year UK
Bond futures	2-year US
Bond futures	5-year US
Bond futures	10-year US
Bond futures	30-year US
Currency forwards	AUD/USD
Currency forwards	EUR/USD
Currency forwards	CAD/USD
Currency forwards	JPY/USD
Currency forwards	NOK/USD
Currency forwards	NZD/USD
Currency forwards	SEK/USD
Currency forwards	CHF/USD
Currency forwards	GBP/USD
Currency forwards	USD Index

Table 1: List of instruments by asset class

Asset Class	Instrument
Commodity futures	ALUMINUM
Commodity futures	NICKEL
Commodity futures	UNLEADED
Commodity futures	ZINC
Equity index futures	ASX SPI 200
Equity index futures	FTSE/MIB
Equity index futures	S&P 500
Bond futures	5-year EURO
Bond futures	30-year EURO
Bond futures	10-year UK

Table 2: List of instruments dropped by asset class

	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
Strategy Performance	0.96	6.39	0.15

Table 3: Performance metrics of the TSMOM

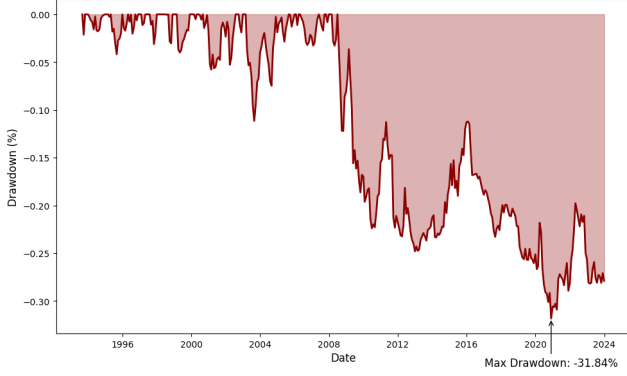


Figure 1: TSMOM portfolio drawdowns

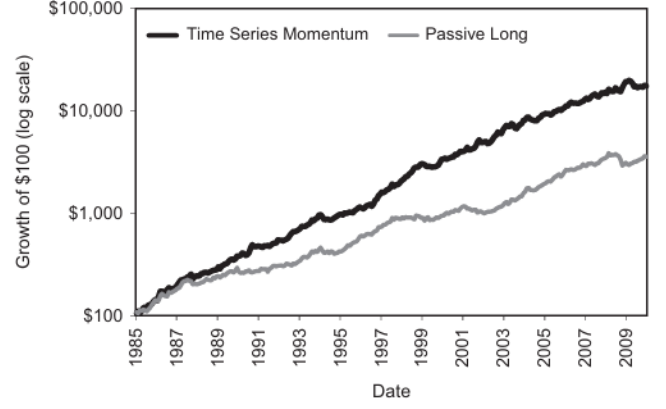


Figure 2: TSMOM portfolio and passive long portfolio cumulative growth of \$100 from Moskowitz, Ooi, and Pedersen (2011)

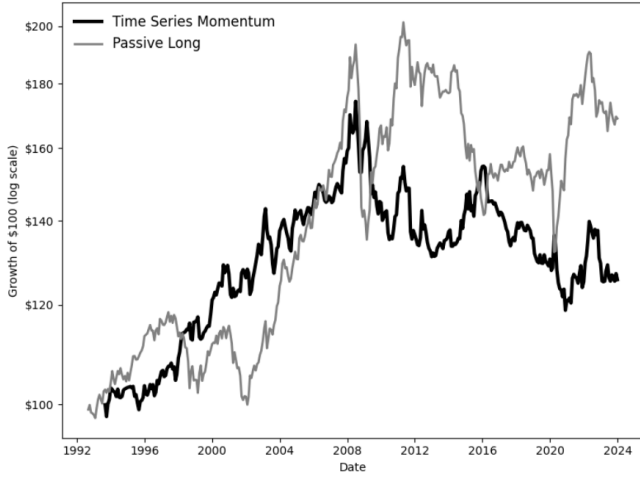


Figure 3: TSMOM portfolio and passive long portfolio cumulative growth of \$100 from our replication

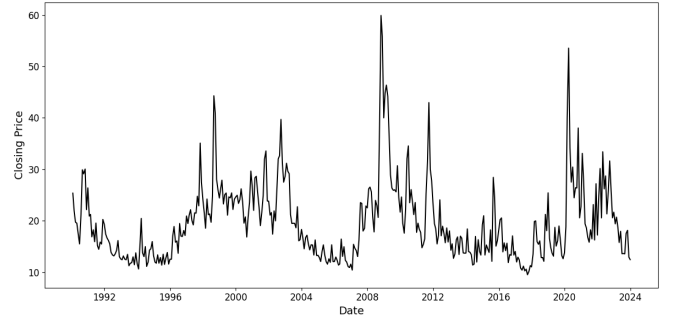


Figure 4: VIX Index

Variable	ADF Statistic	p-value	Critical Value 5%	Stationary (5%)
TSMOM	-9.208888	1.910388e-15	-2.880947	True
VIX	-3.477845	8.574654e-03	-2.880693	True
dVIX	-9.023388	5.678408e-15	-2.880793	True
logVIX	-2.899824	3.498128e-02	-2.880693	True
dlogVIX	-11.002872	6.821820e-20	-2.880670	True

Table 4: ADF test results for stationarity (5% significance level)

Lag	AIC	BIC	FPE	HQIC
0	-10.08	-10.05	4.208e-05	-10.07
1	-11.21	-11.14*	1.306e-05	-11.18*
2	-11.21*	-11.10	1.353e-05	-11.17
3	-11.20	-11.04	1.369e-05	-11.14
4	-11.20	-11.00	1.370e-05	-11.12
5	-11.21	-10.97	1.335e-05	-11.11
6	-11.19	-10.90	1.358e-05	-11.08
7	-11.17	-10.83	1.359e-05	-11.07
8	-11.19	-10.82	1.323e-05	-11.04
9	-11.20	-10.79	1.360e-05	-11.04
10	-11.20	-10.77	1.372e-05	-11.04
11	-11.18	-10.87	1.402e-05	-10.97
12	-11.17	-10.82	1.410e-05	-10.95

Table 5: Lag order selection criteria (lowest values highlighted with *)

```

Summary of Regression Results
=====
Model:                VAR
Method:               OLS
Date:                 Wed, 21, May, 2025
Time:                 23:14:51
=====
No. of Equations:     2.00000    BIC:                -11.1575
Nobs:                 362.000    HQIC:              -11.1963
Log likelihood:       1009.86    FPE:               1.33773e-05
AIC:                  -11.2220    Det(Omega_mle):   1.31583e-05
=====
Results for equation TSMOM
=====
              coefficient      std. error      t-stat      prob
-----
const         0.008135         0.008228         0.989       0.323
L1.TSMOM       0.048588         0.052583         0.924       0.355
L1.logVIX      -0.002518         0.002788        -0.903       0.367
=====
Results for equation logVIX
=====
              coefficient      std. error      t-stat      prob
-----
const         0.514779         0.087918         5.855       0.000
L1.TSMOM       0.463011         0.561873         0.824       0.410
L1.logVIX      0.824276         0.029790        27.670     0.000
=====
Correlation matrix of residuals
              TSMOM    logVIX
TSMOM        1.000000  0.066385
logVIX        0.066385  1.000000

```

Figure 5: OLS Regression Results

Statistic	Value
LM Statistic	22.1934
LM p-value	0.0354
F Statistic	1.8940
F p-value	0.0340

Table 6: Breusch-Godfrey LM Test for Residual Autocorrelation

Eigenvalue	Stability
0.04766	True
0.22348	

Table 7: Eigenvalues of the VAR(1) model – all within unit circle

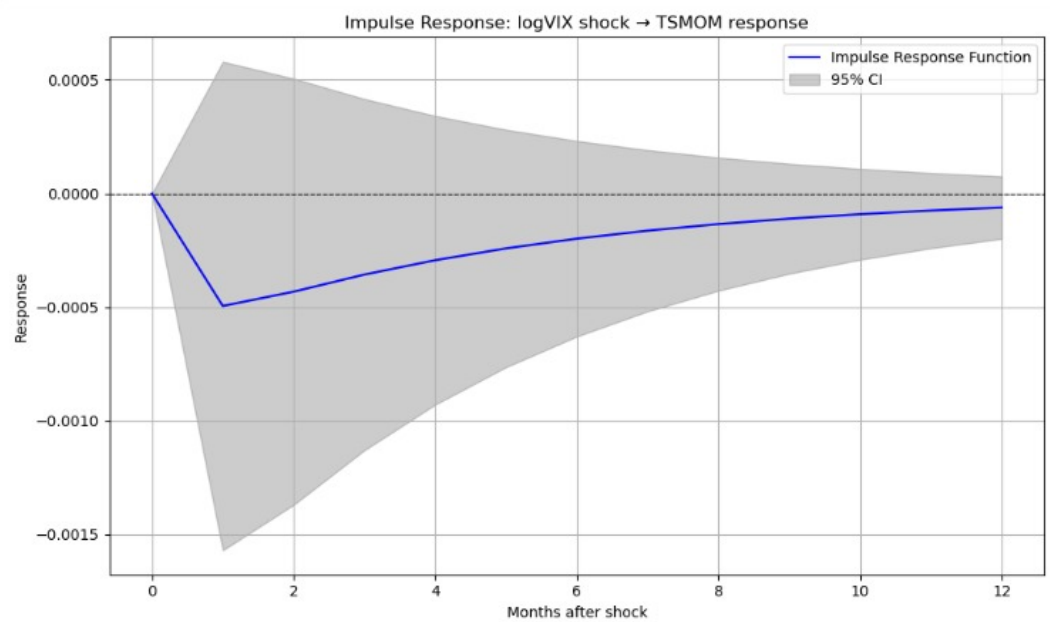


Figure 6: Impulse Response Function

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